Infinitely Many Integer Solutions of $y^2 + x^2y + xz^2 + x + 1 = 0$

Abstract

We give a completely explicit infinite family of integer solutions to

$$y^2 + x^2y + xz^2 + x + 1 = 0.$$

The construction fixes $x = -5$, converts the remaining equation into the norm equation $U^2-5V^2 = 641$, and iterates a single integral norm-preserving transformation. Every algebraic and integrality assertion is proved directly; no general theorem on Pell equations is required.

1 The infinite family

Let

$$\mathcal{F}(x, y, z) = y^2 + x^2y + xz^2 + x + 1.$$

Theorem 1. *Define integer sequences $(y_n)_{n\geq 0}$ and $(z_n)_{n\geq 0}$ by*

$$y_0 = 3, \qquad z_0 = 4,$$

and

$$\boxed{\begin{aligned} y_{n+1} &= 9y_n + 20z_n + 100, \\ z_{n+1} &= 4y_n + 9z_n + 50. \end{aligned}} \tag{1}$$

Then, for every $n \geq 0$ and every $\varepsilon \in \{-1, 1\}$,

$$\boxed{(x, y, z) = (-5, y_n, \varepsilon z_n)} \tag{2}$$

is an integer solution of

$$y^2 + x^2y + xz^2 + x + 1 = 0.$$

All the triples in (2) are pairwise distinct. In particular, the equation has infinitely many integer solutions.

The proof rests on two elementary lemmas.

Lemma 1. *For integers y, z, one has*

$$\mathcal{F}(-5, y, z) = 0 \quad\Longleftrightarrow\quad (2y + 25)^2 - 5(2z)^2 = 641.$$

Thus any solution $(U, V) \in \mathbb{Z}^2$ of

$$U^2 - 5V^2 = 641 \tag{3}$$

with U odd and V even gives an integer solution of the original equation through

$$x = -5, \qquad y = \frac{U - 25}{2}, \qquad z = \frac{V}{2}. \tag{4}$$

Proof. At $x = -5$ the original equation is

$$y^2 + 25y - 5z^2 - 4 = 0.$$

Multiplying by 4 and completing the square gives

$$4y^2 + 100y - 20z^2 - 16 = 0 \quad \Longleftrightarrow \quad (2y + 25)^2 - 20z^2 = 641,$$

which is the asserted equivalence. If U is odd and V is even, the quantities in (4) are integers; substituting $U = 2y + 25$ and $V = 2z$ into (3) then recovers $\mathcal{F}(-5, y, z) = 0$ by the already proved equivalence. $\square$

Lemma 2. *Define*

$$T(U, V) = (9U + 20V,\ 4U + 9V). \tag{5}$$

Then:

1. *T preserves the quadratic form $U^2 - 5V^2$;*
2. *if U is odd and V is even, then the two coordinates of $T(U, V)$ are, respectively, odd and even;*
3. *if $U, V > 0$, then both coordinates of $T(U, V)$ are positive and the first coordinate is strictly larger than U.*

Proof. Writing $T(U, V) = (U', V')$, direct expansion gives

$$\begin{aligned}(U')^2 - 5(V')^2 &= (9U + 20V)^2 - 5(4U + 9V)^2 \\ &= 81U^2 + 360UV + 400V^2 - 80U^2 - 360UV - 405V^2 \\ &= U^2 - 5V^2.\end{aligned}$$

This proves the first assertion. If U is odd and V is even, then $9U + 20V$ is odd and $4U + 9V$ is even, proving the second. Finally, when $U, V > 0$, both new coordinates are positive and

$$(9U + 20V) - U = 8U + 20V > 0,$$

which proves the third assertion. $\square$

Proof of Theorem 1. The pair

$$(U_0, V_0) = (31, 8)$$

satisfies

$$31^2 - 5 \cdot 8^2 = 961 - 320 = 641.$$

For $n \geq 0$, define

$$(U_{n+1}, V_{n+1}) = T(U_n, V_n). \tag{6}$$

By Lemma 2(1), every (U_n, V_n) satisfies $U_n^2 - 5V_n^2 = 641$. Since U_0 is odd and V_0 is even, Lemma 2(2), applied inductively, shows that every U_n is odd and every V_n is even. Hence

$$y_n = \frac{U_n - 25}{2}, \qquad z_n = \frac{V_n}{2}$$

are integers, and Lemma 1 gives

$$\mathcal{F}(-5, y_n, z_n) = 0.$$

Because $\mathcal{F}$ contains z only through z^2, it also follows that $\mathcal{F}(-5, y_n, -z_n) = 0$.

It remains only to verify that these y_n, z_n agree with the recurrence stated in the theorem and that no repetitions occur. From $U_n = 2y_n + 25$ and $V_n = 2z_n$, (6) yields

$$\begin{aligned}y_{n+1} &= \frac{9(2y_n + 25) + 20(2z_n) - 25}{2} = 9y_n + 20z_n + 100, \\ z_{n+1} &= \frac{4(2y_n + 25) + 9(2z_n)}{2} = 4y_n + 9z_n + 50,\end{aligned}$$

which is (1). Also $U_0, V_0 > 0$, so Lemma 2(3) implies inductively that $U_n, V_n > 0$ and $U_{n+1} > U_n$ for every n. Therefore $y_{n+1} > y_n$ and $z_n > 0$. Distinct indices give distinct y-coordinates, while for each fixed index the two choices $\varepsilon = \pm 1$ give distinct z-coordinates. Thus all triples in (2) are pairwise distinct, completing the proof. $\square$

The first four levels of the family are

$$(-5, 3, \pm 4), \quad (-5, 207, \pm 98), \quad (-5, 3923, \pm 1760), \quad (-5, 70607, \pm 31582).$$

Remark 1. Equivalently, the auxiliary sequences admit the compact description

$$U_n + V_n\sqrt{5} = (31 + 8\sqrt{5})(9 + 4\sqrt{5})^n.$$

Indeed, multiplying $U + V\sqrt{5}$ by $9 + 4\sqrt{5}$ produces $(9U + 20V) + (4U + 9V)\sqrt{5}$, which is exactly (6). The recurrence, however, is the entirely integral formulation and is sufficient for the proof.

2 How the construction was found

The term xz^2 suggests first making x negative: this turns the quadratic part in y and z into an indefinite form, the setting in which a single solution can often be propagated. Put $x = -a$ with $a > 0$. Then

$$\mathcal{F}(-a, y, z) = y^2 + a^2 y - az^2 - a + 1, \tag{7}$$

and completing the square gives the general identity

$$\mathcal{F}(-a, y, z) = 0 \quad \Longleftrightarrow \quad (2y + a^2)^2 - a(2z)^2 = a^4 + 4a - 4. \tag{8}$$

Thus a nonsquare value of a would lead to a Pell-type norm equation, provided that one could first locate an integral point.

The dominant terms $a^2 y$ and $-az^2$ cancel when y and z are both of size a. To search systematically near that balance, write

$$y = a - r, \qquad z = a - s,$$

where r, s are small integers. Substitution into (7) gives

$$\mathcal{F}(-a, a - r, a - s) = (1 - r + 2s)a^2 - (2r + s^2 + 1)a + r^2 + 1. \tag{9}$$

The small offsets $(r, s) = (2, 1)$ are especially fruitful:

$$\mathcal{F}(-a, a - 2, a - 1) = a^2 - 6a + 5 = (a - 1)(a - 5).$$

The root $a = 1$ gives the unhelpful seed $(-1, -1, 0)$ within this ansatz. The other root gives

$$a = 5, \qquad (x, y, z) = (-5, 3, 4),$$

and, crucially, the nonsquare coefficient 5 needed for an indefinite norm equation.

For $a = 5$, equation (8) becomes $U^2 - 5V^2 = 641$. The next task is to find an integral linear transformation preserving $U^2 - 5V^2$. A transformation of the standard form

$$(U, V) \longmapsto (pU + 5qV,\ qU + pV)$$

preserves this form whenever $p^2 - 5q^2 = 1$, as a direct expansion shows. The small identity

$$9^2 - 5 \cdot 4^2 = 1$$

therefore supplies $(p, q) = (9, 4)$ and yields precisely

$$(U, V) \longmapsto (9U + 20V,\ 4U + 9V).$$

It preserves the required parity class, maps positive pairs to larger positive pairs, and therefore generates an infinite orbit from $(31, 8)$. This explains both the choice $x = -5$ and the recurrence in Theorem 1 without invoking any external classification theorem.